Abstract Algebra by Pinter, Chapter 20

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Chapter 20 on Integral Domains

Contents

A. Characteristic of an Integral Domain **2**
Q1 . . . 2
Q2 . . . 2
Q3 . . . 2
Q4 . . . 2
Q5 . . . 2
Q6 . . . 2
Q7 . . . 3

B. Characteristic of a Finite Integral Domain **3**
Q1 . . . 3
Q2 . . . 3
Q3 . . . 3
Q4 . . . 3
Q5 . . . 3

C. Finite Rings **4**
Q1 . . . 4
Q2 . . . 4
Q3 . . . 4

D. Field of Quotients of an Integral Domain **4**
Q1 . . . 4
Q2 . . . 5
Q3 . . . 5
Q4 . . . 5
Q5 . . . 5
Q6 . . . 5
Q7 . . . 6

E. Further Properties of the Characteristic of an Integral Domain **6**
Q1 . . . 6
Q2 . . . 6
Q3 . . . 6
Q4 . . . 6
Q5 . . . 6
Q6 . . . 6
Q7 . . . 7

F. Finite Fields **7**
Q1 . . . 7
Q2 . . . 7

A. Characteristic of an Integral Domain

Q1

Let a be any nonzero element of A. If $n \cdot a = 0$, where $n \neq 0$, then n is a multiple of the characteristic of A.

$$n \cdot a = 0 \implies n \cdot 1 = 0$$

but $\text{char}(A) \cdot 1 = 0$, so n is a multiple of the characteristic of A.

Q2

If A has characteristic zero, $n \neq 0$, and $n \cdot a = 0$, then $a = 0$.

$$n \neq 0$$
$$n \cdot a = 0$$
$$n \cdot 1 \cdot a = 0$$

but $n \cdot 1 \neq 0$ because characteristic is zero.

$$a = 0$$

by cancellation property.

Q3

If A has characteristic 3, and $5 \cdot a = 0$, then $a = 0$.

$$3 \cdot a = 0$$
$$5 \cdot a = 3 \cdot a + 2 \cdot a = 2 \cdot a$$
$$2 \cdot a = 0 \implies a = 0$$

Q4

If there is a nonzero element a in A such that $256 \cdot a = 0$, then A has characteristic 2.

$$256 = 2^8$$

characteristic is prime.

$$\text{char}(A) = 2$$

Q5

If there are distinct nonzero elements a and b in A such that $125 \cdot a = 125 \cdot b$, then A has characteristic

$$125 \cdot a = 125 \cdot b$$
$$5 \times 5 \cdot 1 \cdot (a - b) = 0$$
$$a \neq b \implies a - b \neq 0$$
$$5 \cdot 1 = 0$$
$$\text{char}(A) = 5$$

Q6

If there are nonzero elements a and in A such that $(a + b)^2 = a^2 + b^2$, then A has characteristic 2.

Theorem 3: $(a + b)^p = a^p + b^p$

$$(a + b)^2 = a^2 + b^2$$
$$\text{char}(A) = 2$$

Q7

If there are nonzero elements a and b in A such that $10a = 0$ and $14b = 0$, then A has characteristic 2.

$$2 \times 5 \cdot 1 \cdot a = 0$$
$$2 \times 7 \cdot 1 \cdot a = 0$$
$$2 \cdot 1 \cdot (5a + 7a) = 0$$
$$\text{char}(A) = 2$$

B. Characteristic of a Finite Integral Domain

Q1

If A has characteristic q, then q is a divisor of the order of A.

By Lagrange's theorem, any subgroup divides the group order.

$1 + \cdots + 1 = 0$ and so forms a subgroup which divides the group order. The order of this subgroup is the characteristic of A.

See this question.

Q2

If the order of A is a prime number p, then the characteristic of A must be equal to p.

From above the characteristic must divide the order, but the order is prime so the characteristic must equal p.

Q3

If the order of A is p^m , where p is a prime, the characteristic of A must be equal to p.

The characteristic is prime and divides the order, hence $\text{char}(A) = p$.

Q4

$81 = 3 \times 3 \times 3 \times 3$, so by above $\text{char}(A) = 3$.

Q5

If A, with addition alone, is a cyclic group, the order of A is a prime number.

$$A = \langle 1, + \rangle$$
$$\text{ord}(1) = |A|$$
$$\text{ord}(1) \cdot 1 = 0 = |A| \cdot 1$$

but

$$\text{char}(1) \cdot 1 = 0$$

Hence $\text{char}(1) || A|$ But $\text{ord}(1)$ is the smallest n such that $\text{ord}(1) \cdot 1 = 0$ so $\text{ord}(1) | \text{char}(1)$ by Lagrange, and $\text{ord}(1) = |A|$

Thus $|A|$ is prime since $|A| = \text{char}(1)$ which is also prime.

See this question

C. Finite Rings

Q1

Prove every nonzero element of A is either a divisor of zero or invertible.

$$A = \{0, 1, a_1, a_2, \dots, a_n\}$$
$$|A| = n + 2$$

$$a_i 0, a_i 1, a_i a_2, \dots, a_i a_n$$

The size of this subgroup divides $|A|$.

If its size is less than $|A|$, then there exists $a_i x = 0$ where $x \neq 0$. So a_i is a divisor of zero.

Otherwise if its size equals $|A|$, then only $a_i 0 = 0$ and so $a_i x = 1$ meaning a_i is invertible.

Q2

Prove: If $a \neq 0$ is not a divisor of zero, then some positive power of a is equal to 1.

$a \neq 0$ is not a divisor of zero $\implies \text{ord}(a) = |A| \implies A = \langle a \rangle$

But $1 \in A$, so for some x, $a^x = 1$

Q3

If a is invertible, then a^{-1} is equal to a positive power of a.

If a is invertible, $ax = 0 \implies (a^{-1} \cdot a)x = 0 \implies x = 0$.

Therefore a is not a divisor of zero.

$$\text{ord}(a) = |A|$$
$$A = \langle a \rangle$$

So $a^{-1} = a^k$

D. Field of Quotients of an Integral Domain

Q1

$$[a, b] = [r, s] \implies as = br$$
$$[c, d] = [t, u] \implies cu = dt$$

$$[a, b] + [c, d] = [ad + bc, bd]$$
$$[r, s] + [t, u] = [ru + st, su]$$

$$[ad + bc, bd] = [ru + st, su] \implies (ad + bc)su = bd(ru + st)$$
$$adsu + bcsu = bdru + bdst$$
$$as \cdot du + cu \cdot bs = br \cdot du + dt \cdot bs$$

Since $as = br$ and $cu = dt$

$$br \cdot du + dt \cdot bs = br \cdot du + dt \cdot bs$$

So

$$[a, b] + [c, d] = [r, s] + [t, u]$$

Q2

$$[a,b]\cdot[c,d]=[ac,bd]$$
$$[r,s]\cdot[t,u]=[ru,su]$$

$$[ac,bd]=[rt,su] \implies acsu=bdrt$$
$$as\cdot cu=br\cdot dt$$

But $as=br$ and $cu=dt$ Thus

$$[a,b]\cdot[c,d]=[r,s]\cdot[t,u]$$

Q3

$$(u,v)\sim(a,b) \text{ and } (u,v)\sim(c,d) \implies (a,b)\sim(c,d)$$

$$(u,v)\sim(a,b) \implies av=bu$$
$$(u,v)\sim(c,d) \implies cv=du$$

$$v=c^{-1}du$$
$$av=ac^{-1}du=bu$$

$$ad=bc$$
$$(a,b)\sim(c,d)$$

Q4

$$\begin{aligned}[a,b]+([c,d]+[e,f])&=[a,b]+[cf+de,df]\\&=[adf+bcf+bde,bdf]\\([a,b]+[c,d])+[e,f]&=[ad+bc,bd]+[e,f]\\&=[adf+bcf+bde,bdf]\end{aligned}$$

$$[a,b]+[c,d]=[c,d]+[a,b]$$

Q5

$$\begin{aligned}[a,b]\cdot([c,d]\cdot[e,f])&=[a,b]\cdot[ce,df]\\&=[ace,bdf]\\([a,b]\cdot[c,d])\cdot[e,f]&=[ac,bd]\cdot[e,f]\\&=[ace,bdf]\end{aligned}$$

$$[a,b]\cdot[c,d]=[c,d]\cdot[a,b]$$

Q6

$$\begin{aligned}[a,b]\cdot([c,d]+[e,f])&=[a,b]\cdot[cf+de,df]\\&=[acf+bde,bdf]\\[a,b]\cdot[c,d]+[a,b]\cdot[e,f]&=[ac,bd]+[ae,bf]\\&=[acbf+bdae,bdbf]\end{aligned}$$

Both are equivalent so distributive.

Q7

$$\phi(a) = [a, 1]$$

$$\begin{aligned}\phi(ab) &= [ab, 1] \\ &= \phi(a)\phi(b) = [a, 1] \cdot [b, 1] = [ab, 1]\end{aligned}$$

$$\begin{aligned}\phi(a+b) &= [a+b, 1] \\ &= \phi(a) + \phi(b) = [a, 1] + [b, 1] \\ &= [a+b, 1]\end{aligned}$$

E. Further Properties of the Characteristic of an Integral Domain

Q1

$$p \cdot a = 0$$
$$n = p \cdot m + r$$

$$\begin{aligned}n \cdot a &= pm \cdot a + r \cdot a \\ &= m(p \cdot a) + r \cdot a \\ &= r \cdot a\end{aligned}$$

but $n \cdot a = 0$ and $r \neq 0$ because p does not divide n

$$r \cdot a = 0 \implies a = 0$$

Q2

Characteristic is prime and since $a \neq 0$ then the characteristic must be p.

Q3

p^m is a multiple of p. So the characteristic is p.

Q4

$$f(ab) = a^p b^p = f(a)f(b)$$
$$f(a+b) = (a+b)^p = a^p + b^p = f(a) + f(b)$$

Q5

Order is prime and group is cyclic because

$$A \cong \mathbb{Z}_p$$

For any $a \in A : p \neq 0$

$$A = \langle a \rangle$$

Q6

$$\begin{aligned}(a+b)^{p^2} &= [(a+b)^p]^p = [a^p + b^p]^p \\ &= a^{p^2} + b^{p^2}\end{aligned}$$

Assume true for $n = k$

$$\begin{aligned}(a+b)^{p^k} &= a^{p^k} + b^{p^k} \\ (a+b)^{p^{k+1}} &= [(a+b)^{p^k}]^p \\ &= [a^{p^k} + b^{p^k}]^p \\ &= a^{p^{k+1}} + b^{p^{k+1}}\end{aligned}$$

$$\begin{aligned}(a_1 + a_2 + \cdots + a_r)^{p^n} &= [(a_1 + a_2 + \cdots) + a_r]^{p^n} \\ &= (a_1 + a_2 + \cdots)^{p^n} + a_r^{p^n} \\ &= (a_1 + a_2 + \cdots)^{p^n} + a_{r-1}^{p^n} + a_r^{p^n} \\ &= a_1^{p^n} + a_2^{p^n} + \cdots + a_r^{p^n}\end{aligned}$$

Q7

$1 \in A$ and $1 \in B$ $n \cdot 1 = 0$ is true both in A and B

$$\implies \text{char}(A) = \text{char}(B)$$

F. Finite Fields

Q1

A finite field has order prime p and so is isomorphic to the cyclic group (see E5).

That is

$$A = \langle 1, + \rangle$$

Since A is finite order, $\text{char}(A) \neq 0$

Q2

$$f(a) = a^p$$

To show injective(onto):

$$f(x) = f(y) \implies x = y$$

From 18F7, the domain of f is a field and so f is injective.

This can also be shown by

$$\begin{aligned}f(a) = a^p = f(b) = b^p \\ \implies a^p - b^p = 0 \\ \implies (a-b)^p = 0 \\ \implies a = b\end{aligned}$$

f is injective. A has p elements, so the image of f has at least p elements. But the image of f is contained in A, so it has at most p elements.

Therefore f is surjective.